

WORK EXPERIENCE

Data scientistSISL Infotech Pvt LtdOct 2025-Jan 2026

- Developed a **production-grade semantic search model (all-MiniLM-L6-v2)** for the **Samarth Portal**, enabling efficient school information retrieval.
- Implemented a **50% similarity threshold**, improving **retrieval accuracy** and **user search experience**. Built and deployed the complete application stack using **FastAPI (backend)** and **React (frontend)** for seamless **production usage**.
- Integrated the pipeline with **PostgreSQL** and **Docker**, ensuring **scalable and reliable deployment**.
- Built and deployed the application using **FastAPI (backend)** and **React (frontend)** for **production usage**
- Evaluated and compared **multiple LLMs on LM Studio** to select the **best-performing model** for real-world queries.
- Developed an **end-to-end LLM-powered chatbot** using **semantic search + LLM reasoning** for conversational access.

Data science InternHindustan Times Media (Shine)Apr 2025-Oct 2025

- Consolidated three independent models into a **unified BERT-based hierarchical text classification system** (Industry, Department, Role) using **cosine similarity** and **embedding-based label optimization**.
- Designed and implemented **end-to-end NLP pipelines** including **data preprocessing, feature engineering, embedding generation, and multi-level taxonomy classification**.
- Fine-tuned transformer-based architectures**, achieving **89% test accuracy** in **multi-level text categorization** (multi-class classification) across **large-scale structured and unstructured datasets**.
- Developed an **embedding-driven semantic similarity model** to enhance **document matching relevance** and **recommendation precision**.
- Built a **DistilBERT-based document classification system** to structure unstructured text data, achieving **87% validation accuracy**.
- Implemented **Named Entity Recognition (NER) models** for **structured information extraction**, achieving **84% F1-score** on entity-level evaluation.
- Developed and deployed **scalable RESTful APIs using FastAPI**, enabling **real-time inference** and **production-grade model serving**.
- Delivered **end-to-end Machine Learning solutions** including **model training, evaluation, hyperparameter tuning, optimization, and cloud-ready deployment**, improving **automation efficiency** and **reducing manual processing**.

EDUCATION

B.Tech CSE (Data Science)Galgotias University, Greater NoidaOct 2021-Jul 2025
Intermediate (CBSE)Father Agnel School, Greater NoidaMar 2020-Jul 2021

PROJECTS

- Chat Analyzer (NLP)**
Developed an **NLP-based sentiment and topic classification system** using **Python** and **Scikit-learn**, including **text preprocessing, TF-IDF feature extraction, and model evaluation**, achieving **85% accuracy** on labeled chat data.
- Intelligent Document Understanding and Knowledge Extraction Platform**
Designed and deployed an **AI-powered document understanding and information extraction system** using **LLMs (LLaMA via Ollama)**, **transformer-based NLP models**, and a **multi-agent architecture**, where **task-specific agents** handled **document parsing, semantic extraction, entity linking, and contextual reasoning**, enabling **real-time knowledge retrieval and analytics** through **scalable REST APIs**.
- Azure Databricks Real-Time ETL Pipeline**
Developed **scalable batch and real-time data pipelines** using **Azure Databricks, PySpark, Delta Live Tables (DLT), and Spark Structured Streaming** for analytics-ready data. Designed **dimensional data models (Star Schema)** and implemented **Slowly Changing Dimensions (SCD Type 1 & Type 2)** using **Delta Lake** for historical data management. Applied **ETL/ELT processing, data quality checks, partitioning, and performance optimization**, enabling **reliable, production-grade data engineering workflows**.

TECHNICAL SKILLS

- Python, SQL, Machine Learning, Statistical Modeling, Feature Engineering, Model Evaluation** | NumPy, Pandas, Scikit-learn
- Deep Learning & NLP:** PyTorch, TensorFlow, Hugging Face, LLMs, Fine-Tuning, RAG Pipelines, Embeddings, Prompt Engineering
- Data Engineering & Deployment:** PostgreSQL, ETL Pipelines, Data Modeling, PySpark, Databricks, Airflow, Snowflake, Kubernetes, Data Warehousing, Docker, FastAPI, React, AWS, GCP